



A dual-weighted loss function for lightning nowcasting

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ABSTRACT

Lightning associated with severe convective weather poses a significant threat to public safety and infrastructure. While traditional numerical weather prediction and statistical methods have limitations in computational efficiency and accuracy, deep learning (DL) approaches often yield poor predictive performance for rare but critical lightning events due to the severe class imbalance inherent in the data. To address this challenge, this study introduces a dual-weighted cross-entropy loss function (DWCELoss). This novel function combines a static, global class weight with a dynamic, sample-specific grid weight to enhance the model's sensitivity in lightning-prone regions. Experiments on the test set demonstrate that, compared to using class-weight alone, the DWCELoss-trained model (at a 0.9 probability threshold) increases the probability of detection by 78.74% (from 0.334 to 0.597), the critical success index by 36.80% (from 0.299 to 0.409), and the F1-score by 26.30% (from 0.460 to 0.581). A case study further validates the method's ability to capture localized lightning activity, with the CSI score increasing from 0.29 to 0.43 for the event. This work provides a novel and effective pathway for DL-based lightning nowcasting, with significant implications for disaster prevention and public safety.

摘要

闪电活动对公共安全和基础设施构成重大威胁。传统预报方法在效率与精度上存在局限，而深度学习在处理样本极度不平衡的稀有闪电事件时效果欠佳。本研究针对这一挑战，提出了一种创新的双重加权交叉熵损失函数 (DWCELoss)，通过结合静态全局类别权重与动态样本网格权重，显著增强了模型对闪电高发区域的敏感度。实验结果表明，在0.9的概率阈值下，DWCELoss模型相比仅使用类别权重的模型，其检测概率提升78.74%，临近预报成功率提升36.80%，F1分数增长26.30%。本研究为基于深度学习的闪电临近预报开辟了新路径，对防灾减灾具有重要意义。

1. Introduction

Lightning, a hazardous feature of severe convection, threatens public safety, infrastructure, and ecosystems (Cooper and Holle, 2019; Holle, 2016; Veraverbeke et al., 2017). Therefore, accurate lightning nowcasting is crucial for disaster prevention. Traditional methods, including extrapolation techniques (Dixon and Wiener, 1993; Srivastava et al., 2022) and numerical weather prediction models (Fierro et al., 2013; Lynn and Yair, 2010), face limitations in capturing rapid storm evolution or entail high computational costs (Rojas-Campos et al., 2023; Serifi et al., 2021).

In recent years, deep learning (DL) has advanced rapidly, excelling in handling high-dimensional meteorological data. It has emerged as a powerful tool for meteorological forecasting, demonstrating success in predicting various phenomena (Badrinath et al., 2023; Chkeir et al., 2023; Farmanifard et al., 2023; Fister et al., 2023; Kolios, 2023; Salcedo-Sanz et al., 2024; Zhou et al., 2019) and showing promise for lightning nowcasting. For example, Lin et al. (2019) incorporated an attention mechanism into a dual-source spatiotemporal neural network (ADSNet), enhancing key feature extraction and improving predictive accuracy. Zhou et al. (2020) developed LightningNet, integrating satellite, radar, and lightning data, achieving high performance in 0–1 h lightning now-

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casting. Leinonen et al. (2022) designed a recurrent convolutional network to predict lightning probabilities within 60 min at a 5-min resolution.

However, lightning is strongly linked to convective systems (Deierling and Williams, 2016; Liu et al., 2010; Utsav et al., 2022; Yi et al., 2013) and predominantly occurs at the meso- γ scale (Orlanski, 1975). Spatially, lightning regions are much smaller than non-lightning regions, exhibiting significant class imbalance (Leinonen et al., 2022), where lightning (positive class) is vastly outnumbered by non-lightning (negative class). Similar to the challenges faced in gale forecasting (Liang and Hu, 2022), the significant class imbalance in lightning data skews models toward the majority class, significantly impairing performance. Common mitigation strategies like downsampling (Lee and Seo, 2022), undersampling (Tyagi and Mittal, 2020), oversampling (Zhu et al., 2021), and ensemble learning (Liu et al., 2017) can introduce bias or information loss.

In DL, the loss function directly governs model learning. While adjusting loss functions is an effective way to handle class imbalance (Mansouri et al., 2023; Salehi et al., 2017; Yeung et al., 2022; You et al., 2023), existing approaches for lightning forecasting remain limited. The conventional weighted cross-entropy loss (WCELoss) uses static class weights to address global imbalance but suffers from two key shortcomings: (1) the class weights are fixed during training, unable to dynamically adjust for individual cases with different balance ratio (e.g., localized lightning events), and (2) current implementations rely solely on a single physical feature, failing to integrate additional empirical constraints (e.g., lightning frequency patterns) for lightning-prone regions.

To overcome these limitations, we propose a dual-weighted cross-entropy loss function (DWCELoss)—a static class weight to counter global imbalance with a dynamic, sample-specific grid weight derived from localized lightning frequency. This strategy achieves dual optimization—balancing the global lightning/non-lightning distribution while dynamically enhancing sensitivity to high-risk regions. Meanwhile, the grid weight further provides frequency-informed guidance by encoding spatial lightning frequency as prior knowledge. We validate DWCELoss through comprehensive experiments, demonstrating its effectiveness in significantly improving key nowcasting metrics compared to baseline methods.

2. Dataset

The study area is North China (32°–42°N, 110°–120°E), where convection is frequent in summer, accompanied by high lightning incidence. The radar composite reflectivity (CREF) effectively represents the distribution, intensity, and type of precipitation particles, which are physically closely associated with lightning activity. This study integrates CREF and lightning location data to predict lightning in the next 0–1 h. The observational networks comprise 46 operational weather radars and 64 lightning detection stations within this domain. Their spatial distribution is detailed in the supplementary materials (Fig. S1).

2.1. Data sources and preprocessing

The CREF data from the China Meteorological Administration have a temporal resolution of 6 min and a spatial resolution of 0.01°, and represent the intensity and structure of convective precipitation, which is physically closely linked to lightning activity. In this study, reflectivity factor values below 0.1 dBZ were set to zero to filter out weak system noise. To enhance neural network efficiency and match the physical scale of lightning activity, these data were downsampled to a 256 × 256 grid (0.039° resolution) using bilinear interpolation.

The lightning location data, provided by the Institute of Electrical Engineering, Chinese Academy of Sciences, record the time, geographic coordinates (latitude, longitude), altitude, and discharge intensity of lightning in a spatial scatter format, with a temporal resolution of less than 3 ms. Quality control was conducted following established methods

(Meng et al., 2022). Data were gridded to a 256 × 256 (0.039° resolution) spatial grid to align with the radar data, with a temporal resolution of 60 min. Each grid cell records a binary variable (0 for no lightning, 1 for lightning) and the lightning frequency. During training, the binary variable served as the label, while lightning frequency was incorporated into the loss function as a proxy for lightning activity.

2.2. Dataset split

Lightning typically occurs in convective weather and evolves rapidly (Deierling and Williams, 2016). To better capture its evolution, CREF time series were used as input, each consisting of 10 consecutive time steps, spaced 6 min apart, denoted as $\{X_k \mid k = t - i \times \Delta t, i = 0 \sim 9, \Delta t = 6 \text{ min}\}$, where t denotes the current time and Δt denotes the time interval. The corresponding label represents lightning occurrence within 0–1 h, $\{Y_{t \sim t+10\Delta t}\}$, indicated as 0 (no lightning) or 1 (lightning).

The dataset was constructed from preprocessed data from July 2024. The top 2000 samples with the most significant lightning activity were selected (details provided in the supplementary materials). To ensure independence, the dataset was split chronologically into training, validation, and test sets in an 8:1:1 ratio (1600, 200, and 200 samples, respectively).

3. Methods

3.1. Architecture

The model used in this study is built upon the UNet architecture, which is renowned for its effective fusion of high-level semantic features and low-level spatial details via skip connections (Ronneberger et al., 2015). This architecture has demonstrated superior performance in spatiotemporal feature extraction and is widely utilized in the meteorological field (Fister et al., 2023; Lyu et al., 2024; Trebing et al., 2021). Based on UNet, the proposed model employs a classic “U”-shaped structure, comprising a downsampling encoder path, an upsampling decoder path, and the connecting skip connections (Fig. S2). The encoder extracts abstract features by progressively reducing spatial dimensions, while the decoder restores the spatial resolution to generate fine-grained predictions. Details are provided in the supplementary materials.

3.2. Loss function

To address the severe class imbalance in lightning prediction, this study employs a weighted cross-entropy loss function (WCELoss), defined as follows:

$$\text{WCELoss} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=0}^1 w_c y_{i,c} \log(p_{i,c}), \quad i = \{1, 2, \dots, N\}, \quad c = \{0, 1\}, \quad (1)$$

where N represents the total number of samples, i represents the sample index, y denotes the label, p indicates the predicted probability, c denotes the class (0 for non-lightning, 1 for lightning), w_c is the static class weight derived from training set statistics (approximately 1:10 based on flash frequency and 1:65 based on grid counts) to increase the focus on the minority (lightning) class. Detailed derivations are provided in the supplementary materials.

However, a static weight fails to capture local variations in lightning activity. Therefore, we propose an innovative dual-weighted cross-entropy loss function (DWCELoss). This function introduces a dynamic, sample-specific grid weight (gw_i), derived from lightning frequency, in addition to the class weight (w_c). This compels the model to focus on regions with higher lightning incidence. The DWCELoss is formulated as follows:

$$\text{DWCELoss} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=0}^1 w_c y_{i,c} \cdot \log(p_{i,c}) \cdot gw_{i,c}, \quad i = \{1, 2, \dots, N\}, \quad c = \{0, 1\}. \quad (2)$$

The detailed motivation, implementation, and step-by-step derivation for both loss functions are provided in the supplementary materials.

Table 1 Experiments configuration.

Group	Expname	Loss_Function	Class-W ($w_0:w_1$)	Grid-W
Group-1	CE	Cross-Entropy	×	×
	WCE_flash	Weighted Cross-Entropy	1:10	×
	WCE_num	Weighted Cross-Entropy	1:65	×
Group-2	GCE	Grid-Weighted Cross-Entropy	×	✓
	DWCE_flash	Dual-Weighted Cross-Entropy	1:10	✓
	DWCE_num	Dual-Weighted Cross-Entropy	1:65	✓

Table 2 Performance metrics at probability threshold 0.9 for all experiments on TestSet.

Group	Expname	CSI	POD	FAR	F1
Group-1	CE	0.001	0.001	0.000	0.001
	WCE_flash	0.143	0.145	0.081	0.250
	WCE_num	0.299	0.334	0.259	0.460
Group-2	GCE	0.020	0.020	0.063	0.040
	DWCE_flash	0.131	0.134	0.159	0.232
	DWCE_num	0.409	0.597	0.435	0.581

3.3. Evaluation metrics

The model's lightning nowcasting performance was evaluated using four standard meteorological metrics: the probability of detection (POD), false alarm ratio (FAR), critical success index (CSI), and the F1-score.

To ensure high confidence in predictions, a probability threshold of 0.9 was used to classify grid points as lightning events. The evaluation was performed using a neighborhood-based method to account for minor spatial displacements. A forecast was considered a "hit" if an observed lightning event occurred within a $12 \text{ km} \times 12 \text{ km}$ (3×3 grid) neighborhood of the predicted grid point. The detailed definitions and formulas for all metrics are provided in the supplementary materials.

4. Results

To systematically evaluate the proposed loss functions and weighting strategies, we conducted two groups of experiments. Group 1 utilized the conventional WCELoss, while Group 2 was based on our proposed DWCELoss. Each group comprised three sub-experiments with distinct class-weight and grid-weight configurations. The setup for all six experiments is summarized in Table 1, with details provided in the supplementary materials.

4.1. Effects of class weight

The evaluation of class-weighting strategies (Group 1), summarized in Table 2 and visualized in Fig. 1, reveals their significant impact on model performance at a 0.9 probability threshold.

The baseline CE model, without any weighting, failed completely, yielding zero for all metrics (CSI, POD, FAR, F1). This indicates that the model was overwhelmed by the class imbalance, learning only to predict the dominant negative class. In contrast, both weighting strategies, WCE_flash and WCE_num, substantially improved performance, confirming the effectiveness of redirecting the model's focus to the minority class.

A direct comparison shows that the grid count-based weighting (WCE_num) outperformed the frequency-based method (WCE_flash), achieving higher POD and F1 scores with an expected minor increase in FAR. This quantitative superiority is visually corroborated in Fig. 1, where WCE_num's predictions align more closely with observed lightning patterns.

However, despite these gains, the overall POD remained limited, suggesting that the effectiveness of a single, global class-weight is constrained. This limitation necessitates the exploration of more advanced weighting strategies to significantly improve predictive accuracy for minority classes.

4.2. Effects of dual-weight synergy

To address the limitation of global class weights, which overlook sample-specific characteristics, we introduced a dynamic grid weight based on lightning frequency. This acts as a frequency-informed prior, guiding the model to focus on regions with high lightning activity. The effectiveness of this dual-weighting approach was validated in Group 2 experiments (Table 2).

The results indicate that relying solely on grid weight (GCE) yielded minimal improvement (CSI of 0.020). Furthermore, when both class and grid weights were derived from the same physical feature (flash frequency), the DWCE_flash model's performance did not improve over its single-weight counterpart, as redundant information led to a significantly higher FAR.

In stark contrast, a powerful synergistic effect was observed in the DWCE_num experiment. By combining a class weight based on spatial extent (grid count) with a grid weight based on lightning frequency, the model achieved substantial gains: the CSI increased by 36.80% (from 0.299 to 0.409); the F1-score by 26.30% (from 0.460 to 0.581); and the POD by a remarkable 78.74% (from 0.334 to 0.597). While the FAR also increased, the overall balance between hit rate and false alarms was far superior.

These results validate that the proposed DWCELoss is most effective when its dual-weighting mechanism leverages complementary physical features. This strategy achieves the crucial goal of significantly improving POD while suppressing the growth of FAR, thereby providing a more effective guidance mechanism for lightning nowcasting.

4.3. Analysis on a typical lightning event

To further validate the role of the grid weight, a case study was conducted on a lightning event with a forecast valid for the period 0642–0742 UTC 30 July 2024. A qualitative comparison (Fig. 2) shows that while both the WCE_num and DWCE_num models predicted the event's general shape, the WCE_num forecast produced substantially lower probabilities and failed to capture the core of the event in southern Beijing. In contrast, the DWCE_num forecast, guided by the grid-weight map, generated high-probability predictions that aligned well with observed lightning patterns, including both dense and scattered events (e.g., in Shiyan near $(32.1^\circ\text{N}, 110.5^\circ\text{E})$ and southwestern Chengde near $(41.0^\circ\text{N}, 117.0^\circ\text{E})$).

This qualitative superiority is confirmed by quantitative metrics (Fig. 3). For this specific event, DWCE_num achieved a CSI of 0.429, a POD of 0.644, and an F1-score of 0.600. These robust scores significantly outperform the single-weight model and are consistent with our overall statistical conclusions, validating that the dual-weighting strategy effec-

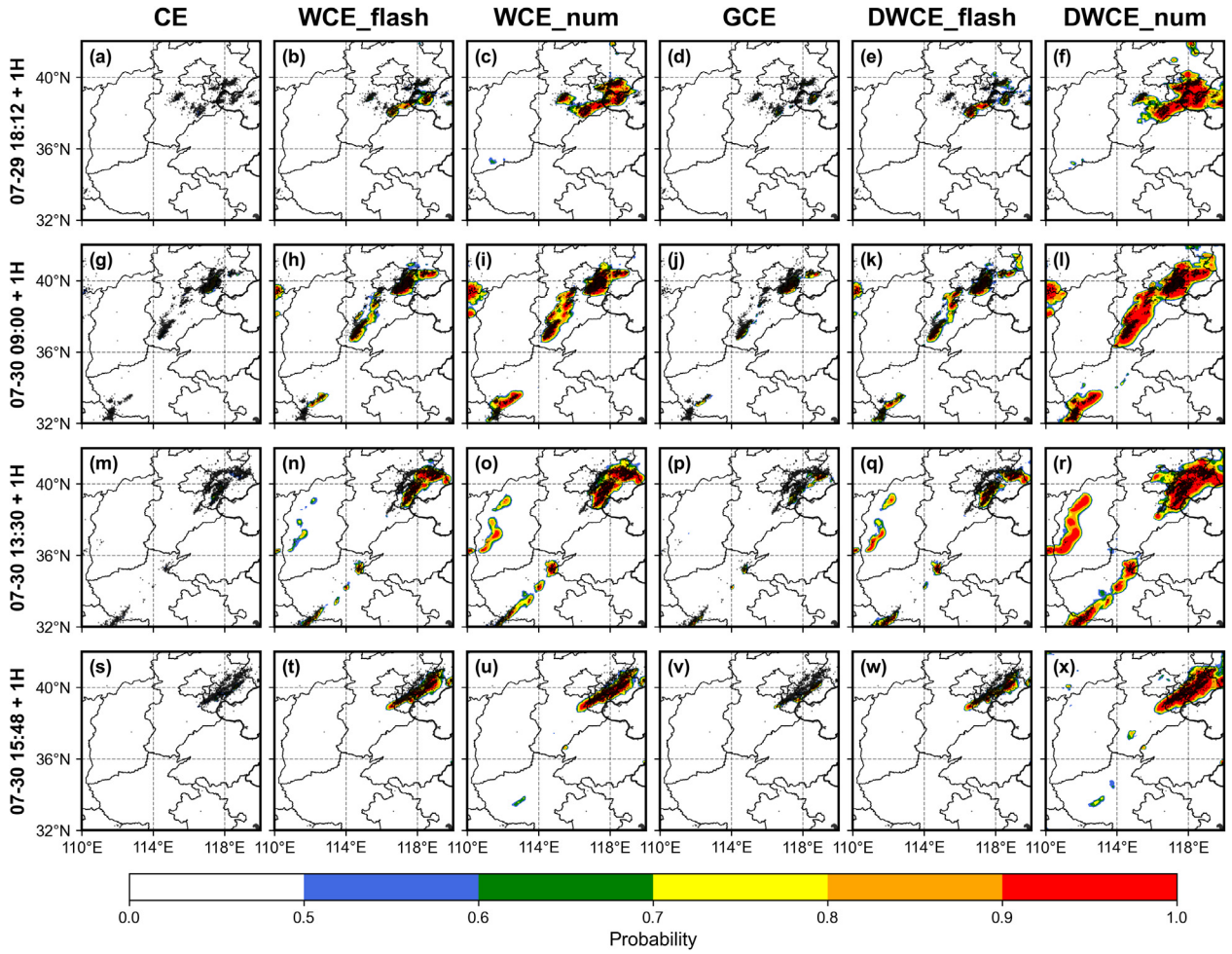


Fig. 1. The 0–1 h lightning probability forecasts from six model experiments for four case studies, issued at (a–f) 1812 UTC 29 July 2024; and (g–x) 0900, 1330, and 1548 UTC 30 July 2024. Filled contours represent the predicted probability, while black markers indicate observed lightning during the corresponding 1 h verification period.

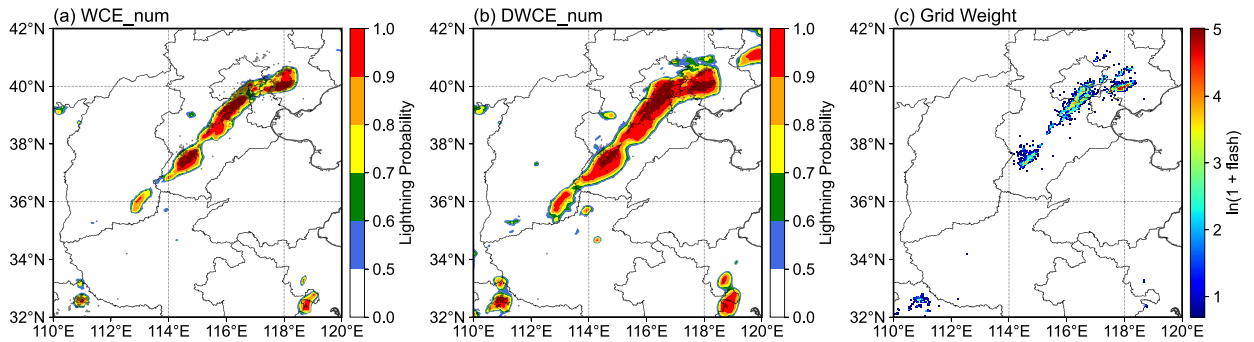


Fig. 2. Forecasts and observations for the 0–1 h period 0642–0742 UTC 30 July 2024. (a, b) Predicted lightning probability (filled contours) from the WCE_num and DWCE_num models, respectively, overlaid with observed lightning locations (black markers). (c) The corresponding grid weight (log-transformed observed flash counts) distribution.

tively improves the model's ability to capture the spatial distribution of lightning and to pinpoint areas of high flash frequency.

4.4. Generalization experiments

To further assess the model's generalization capabilities, the DWCE_num model was tested on two unseen, typical convective events outside the training dataset: a discontinuous linear convective system and an areal convective system.

As shown by the quantitative metrics in Table S1 and the visual forecasts in Fig. S2, the model demonstrated robust performance for both patterns. For the linear system, the model achieved a CSI of 0.433 and a POD of 0.664. For the areal system, the CSI was 0.433 with a POD of 0.592, and the FAR remained below 0.45 for both cases.

These results confirm DWCE_num's strong generalization ability across diverse meteorological scenarios, underscoring its potential for practical forecasting applications. A detailed meteorological description and qualitative analysis of these events are provided in the supplementary materials.

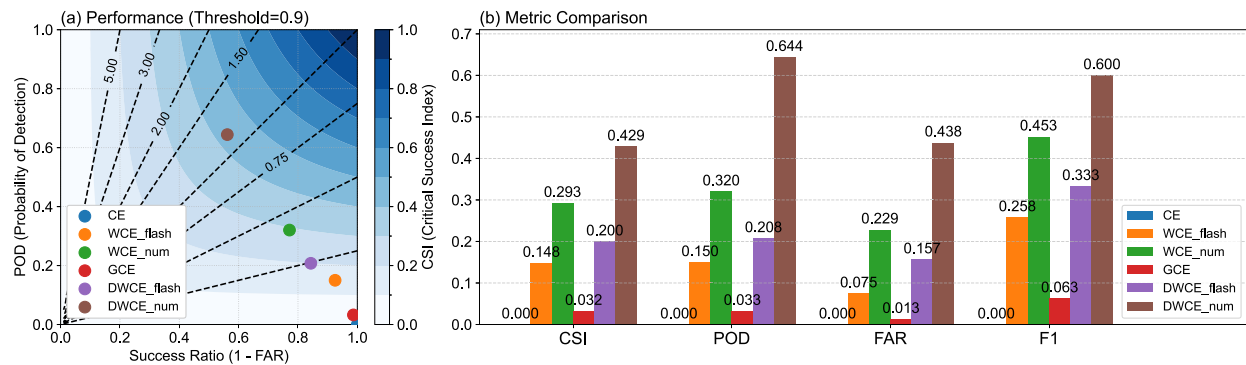


Fig. 3. (a) Performance diagram and (b) bar chart comparing key metrics (CSI, POD, FAR, and F1 Score) across all models. Both panels correspond to the 0–1 h forecast period 0642–0742 UTC 30 July 2024.

5. Discussion and conclusion

This study developed and validated a novel dual-weighted cross-entropy loss function (DWCELoss) to address the severe class imbalance in DL-based lightning nowcasting. DWCELoss synergistically combines a static, global class-weight to counteract the overall sample imbalance with a dynamic, sample-specific grid weight to focus the model on spatially concentrated lightning activity. By embedding lightning frequency as a guiding prior, the loss function more effectively captures the spatiotemporal evolution of lightning.

Through a series of systematic experiments comparing unweighted, single-weighted (WCELoss), and dual-weighted (DWCELoss) approaches, this study demonstrated the superiority of the dual-weighting strategy. The most significant performance gains were achieved when the two weights were derived from complementary physical features: a class weight based on the spatial extent of lightning (grid count) and a grid weight based on its lightning frequency. This configuration achieved the crucial objective of substantially improving the POD while controlling the growth of the FAR. Furthermore, generalization experiments confirmed the model's robustness, showing effective performance on unseen discontinuous linear and areal convective systems, highlighting its potential for practical applications.

To contextualize our results within the broader field, we reference a benchmark study (Zhou et al., 2020) that evaluated a multi-source model (LightningNet) under different data configurations (5 km grid, 20 km tolerance radius). Their reported CSI scores were 0.350 (satellite only), 0.364 (satellite + radar), 0.419 (radar + lightning), 0.432 (satellite + lightning), and 0.453 (all three). While direct numerical comparison is constrained by differences in resolution (4 km vs. 5 km) and spatial tolerance (12 × 12 km neighborhood vs. 20 km radius), our model, using only single-source radar data (CREF) enhanced by DWCELoss, achieved a CSI of 0.409. This indicates that our focus on addressing class imbalance via the loss function allows a streamlined, single-source model to approach the performance level of more complex, multi-source architectures. It underscores the efficacy and potential of the DWCELoss strategy.

Despite these promising results, this study acknowledges several limitations. First, the weighting factors were derived from a limited set of physical features; future research could explore incorporating other characteristics such as lightning intensity and spatial distribution patterns. Second, the model's forecasting performance was found to be sub-optimal for large-scale linear convective systems. This is potentially attributable to the complex dynamics of such systems, inherent limitations in ground-based lightning detection networks (particularly for in-cloud lightning), and insufficient representation of these events in the training data. Third, the dataset is limited to July 2024; while capturing peak summer convection, it requires further validation with longer time series to ensure cross-seasonal generalization. Future work will there-

fore focus on integrating alternative data sources, such as satellite observations, and expanding the training dataset to refine the dual-weight strategy and enhance its applicability to a wider range of severe weather scenarios.

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Supplementary materials

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